

Smart Classroom Monitoring System

Objective:

1. To monitor classroom temperature, light intensity, and motion using sensors.
2. To automatically control fan, light, and buzzer for a smart classroom environment.

Introduction:

A Smart Classroom Monitoring System is an Arduino-based project that automatically monitors classroom conditions. It uses sensors to detect temperature, light level, and motion, then controls classroom devices such as a fan, light, and buzzer. The system also displays information on an LCD.

Required Equipment:

- Arduino Uno
- 16×2 LCD Display
- PIR Sensor
- TMP36 Temperature Sensor
- LDR
- LED
- Buzzer
- DC Motor/Fan
- NPN Transistor
- Resistors
- Breadboard
- Jumper Wires

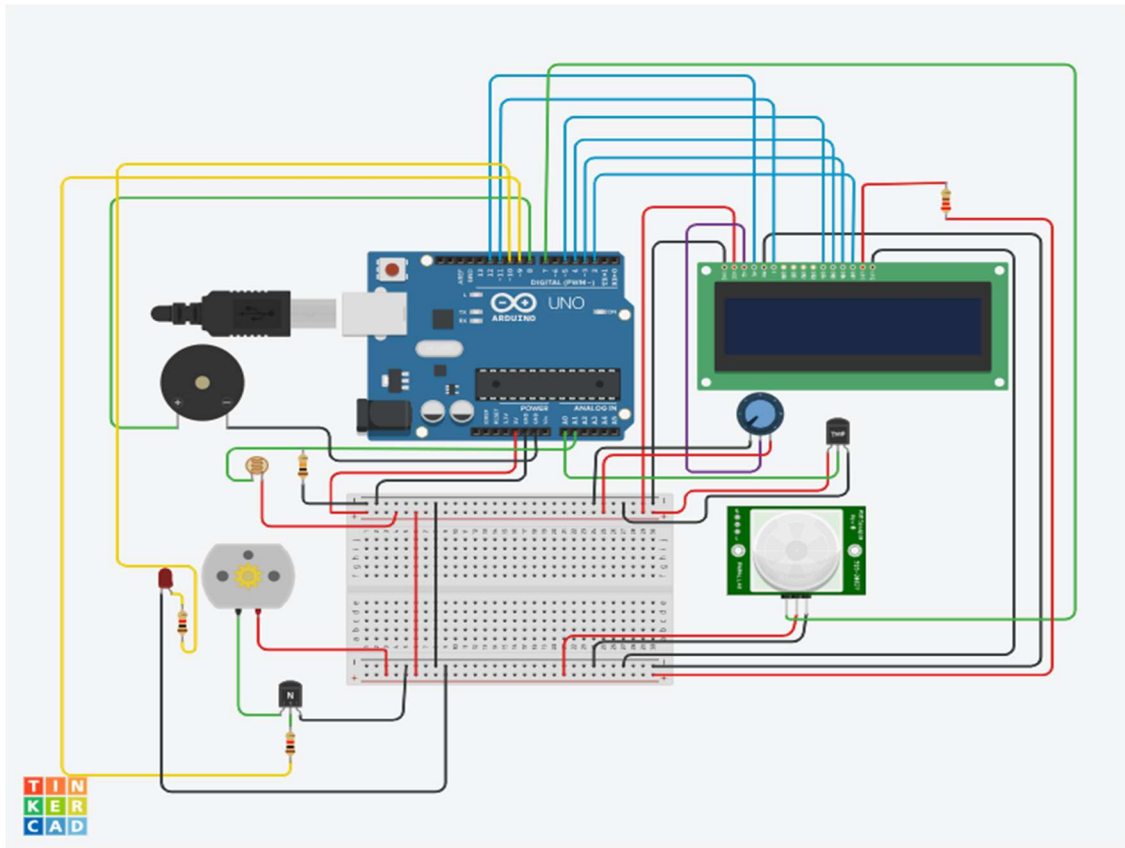


Figure 1: Circuit Diagram of Smart Classroom Monitoring System.

Working Procedure

1. Power ON the Arduino.
2. LCD displays the project name.
3. TMP36 measures temperature.
4. LDR measures light intensity.
5. PIR sensor detects motion.
6. If temperature $> 30^{\circ}\text{C}$, the fan turns ON.
7. If light intensity $< 20\%$, the LED turns ON.
8. When motion is detected, the buzzer turns ON and the motion count increases.
9. LCD shows temperature, light percentage, motion count, and motion status.

Result

- LCD displayed all sensor data correctly.

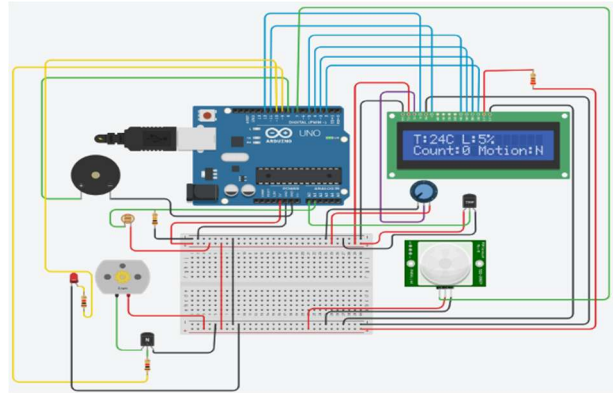


Figure 02 : LCD Display All Data

- Motion was detected using the PIR sensor.

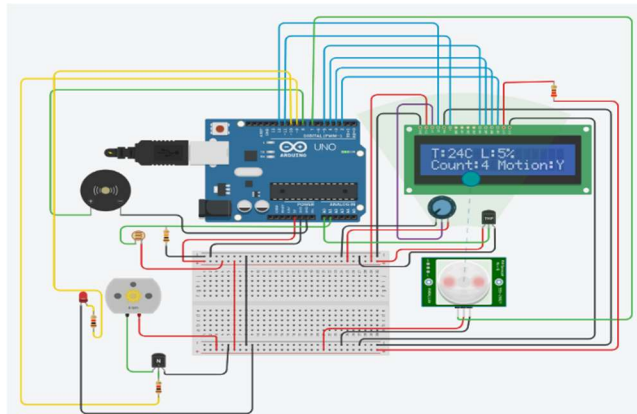


Figure 03 : LCD Display PIR Sensor Data (Motion & Count)

- Temperature was successfully monitored and fan operated automatically.

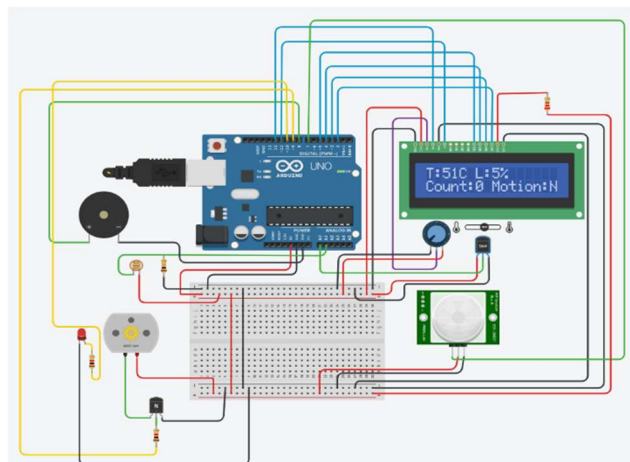


Figure 04 : Temperature Monitoring and Automatic Fan Control Using TMP36 Sensor

- Light intensity was detected and LED controlled automatically.

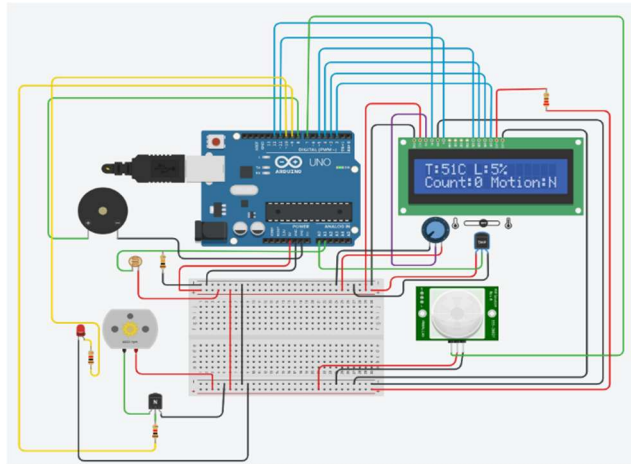


Figure 05 : Automatic Light Control Using LDR and LED

Discussion:

The project successfully automated classroom monitoring using Arduino. It improved energy efficiency by automatically controlling lights and fans based on environmental conditions. The system is simple, low-cost, and suitable for smart classroom applications.